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# **LAB 03 INSTRUCTIONS**

For

## **Create an analyst profile**

Course Code: C1123

Conducted by

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**Version 5.1**

## Lab 03 — Create an analyst profile

**Course:** Wireshark Network Analysis Masterclass (C1123)

**Topic:** 03 — Global Preferences and Troubleshooting Profiles

**Learning outcome:** LO1 — Plan scoped captures and configure a reproducible analyst profile.

**Time:** about 60 minutes

**Version:** v5.1 · 5 October 2026

### Scenario

Three analysts will share findings on the same case. To make results reproducible, you build a named analyst profile with numeric addresses and evidence columns.

### Goal

A profile bundles reproducible columns, coloring rules and protocol settings.

### What you will produce

A reusable C1123-Analyst profile and its folder path.

### Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

### Files for this lab

| File                         | What it is for                                           |
|------------------------------|----------------------------------------------------------|
| data/branch-office.pcap      | Synthetic branch-office capture used by the lab          |
| assets/scenario.md           | The help-desk ticket that sets the scenario              |
| assets/checks.json           | Filters and frame counts the fixture must satisfy        |
| assets/expected-evidence.png | The packet list your filter should produce               |
| scripts/verify.py            | Checks the capture facts with TShark                     |
| scripts/export_evidence.py   | Exports this lab's evidence rows to outputs/evidence.csv |
| outputs/findings.md          | Findings sheet you complete                              |

### Lab at a glance

1. Create the C1123-Analyst profile
2. Turn off name resolution
3. Add stream index and delta columns
4. Record and export the profile folder

## Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Create a profile named C1123-Analyst using Edit > Configuration Profiles. Keep the original profile available.
3. Open data/branch-office.pcap. Disable network name resolution in View > Name Resolution. Record numeric client, server and resolver IP addresses.
4. Select a TCP packet. Expand TCP and right-click Stream index > Apply as Column. Add `frame.time_delta_displayed` as a custom column using Preferences > Appearance > Columns.
5. Apply dns and inspect the profile name and displayed columns. Copy the profile folder path from Help > About Wireshark > Folders into outputs/findings.md. Export or copy your profile to outputs/profile/ without overwriting a colleague profile.
6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in outputs/findings.md.

```
python3 scripts/export_evidence.py
```

## Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

| dns                                |          |            |             |          |                                                      |
|------------------------------------|----------|------------|-------------|----------|------------------------------------------------------|
| No.                                | Time     | Source     | Destination | Protocol | Info                                                 |
| 3                                  | 0.020000 | 192.0.2.10 | 192.0.2.53  | DNS      | Standard query 0x0065 A portal.example.test          |
| 4                                  | 0.040000 | 192.0.2.53 | 192.0.2.10  | DNS      | Standard query response 0x0065 A portal.example.t... |
| 5                                  | 0.050000 | 192.0.2.10 | 192.0.2.53  | DNS      | Standard query 0x0066 A missing.example.test         |
| 6                                  | 0.070000 | 192.0.2.53 | 192.0.2.10  | DNS      | Standard query response 0x0066 No such name A mis... |
| 7                                  | 0.080000 | 192.0.2.10 | 192.0.2.53  | DNS      | Standard query 0x0067 A slow.example.test            |
| 8                                  | 0.880000 | 192.0.2.53 | 192.0.2.10  | DNS      | Standard query response 0x0067 A slow.example.tes... |
| branch-office.pcap    Displayed: 6 |          |            |             |          |                                                      |

*Lab 03 expected evidence*

## Test it

The supplied dns expression matches 6 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in outputs/. Explain the observed result rather than only copying a count.

## Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.

- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching lab-tls.keys file by absolute path, reload, and remove a key-log preference from another lab.

### Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y dns -T fields -e frame.number -e ip.src -e ip.dst -e dns.qry.name
```

### Challenge

Develop and explain an alternative filter.

### Reflection

Which second observation point would strengthen your conclusion?

### Extension (optional)

Export your C1123-Analyst profile folder and import it on a second machine; confirm the columns appear.

### Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

**Note:** The same steps appear in the Learner Guide. The slides show only the scenario and a summary.